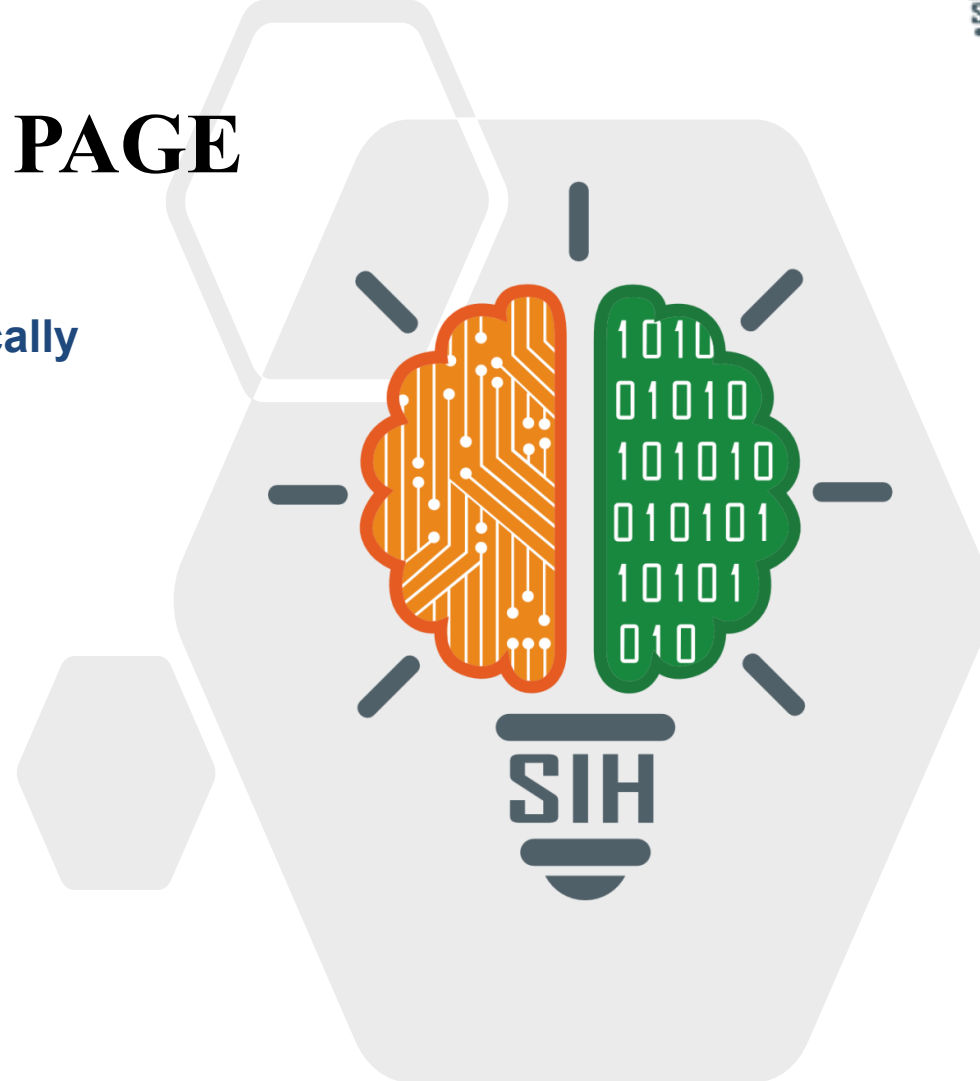


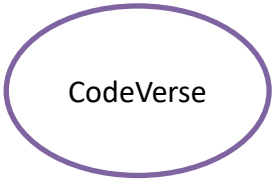
SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – **SIH26250**
- Problem Statement Title- **Planning and dynamically retasking air operations in a contested, rapidly changing environment**
- Theme- **Transportation & Logistics**
- PS Category- **Software**
- Team ID- **133847**
- Team Name (Registered on portal)- **CodeVerse**
- GitHub- **Suyash2527/airpower-ops-optimiser**





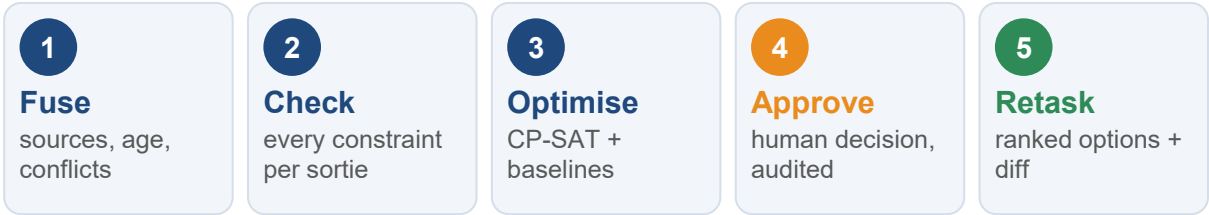
AIRPOWER: AI Decision Support for Air Operations



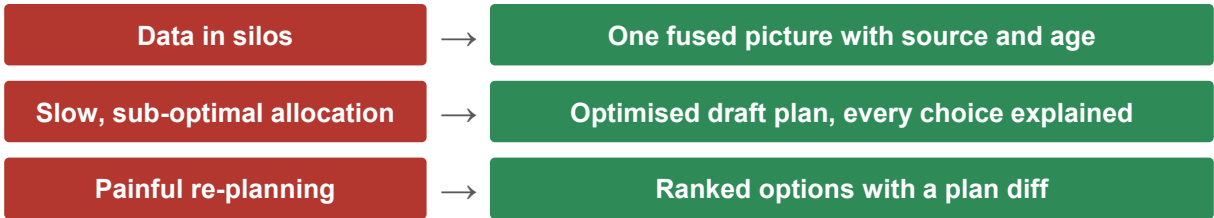
❖ Proposed Solution (Describe your Idea/Solution/Prototype)

AirPower fuses scattered air-operations data into one live picture, builds an optimised plan with a reason for every assignment, and proposes ranked re-plans the moment anything changes. **A human approves every change.**

• Detailed explanation of the proposed solution

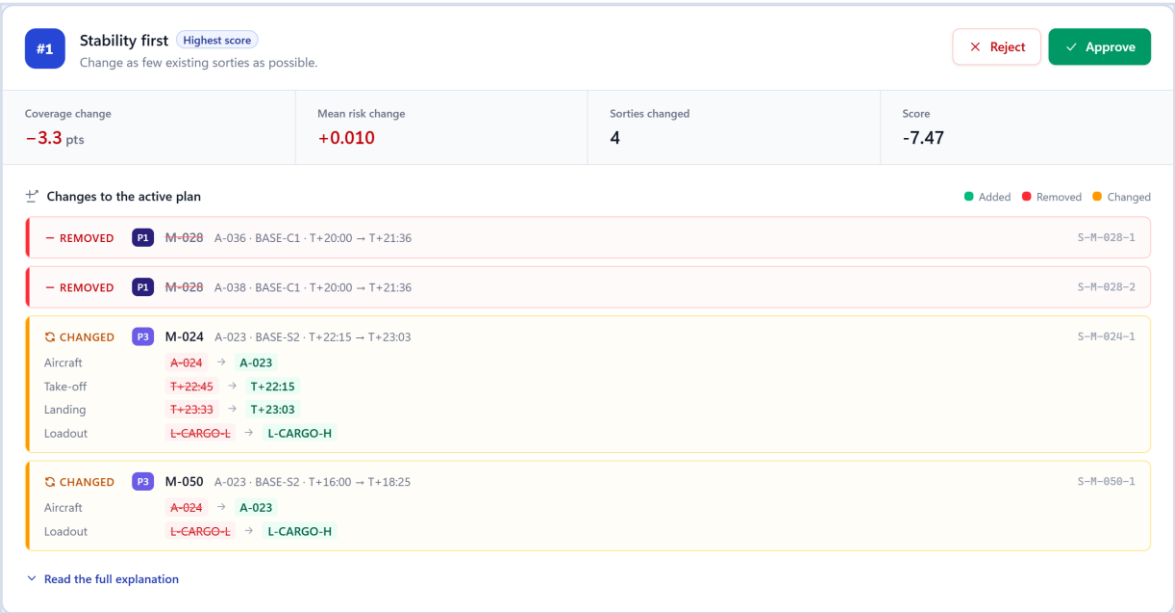


• How it addresses the problem

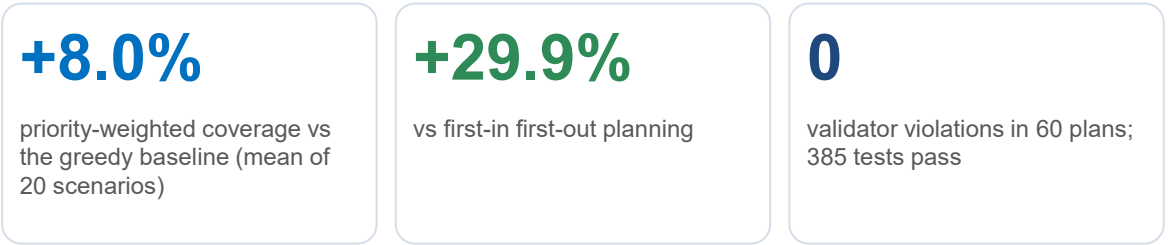


• Innovation and uniqueness of the solution

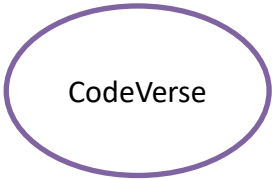
- Explainable by construction: reason code + sentence + next-best option
- Stability-aware retasking: airborne sorties frozen, minimal-change variants
- Independent validator: 0 violations across 60 benchmark plans



Retasking option as the planner sees it: trade-offs and a sortie-level diff (prototype screenshot)



Measured by benchmarks/run_benchmarks.py on synthetic scenarios; see benchmarks/RESULTS.md



TECHNICAL APPROACH



❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

Frontend

Next.js · React · TypeScript · Tailwind. India map drawn in SVG from open data (no tile server).

Optimisation

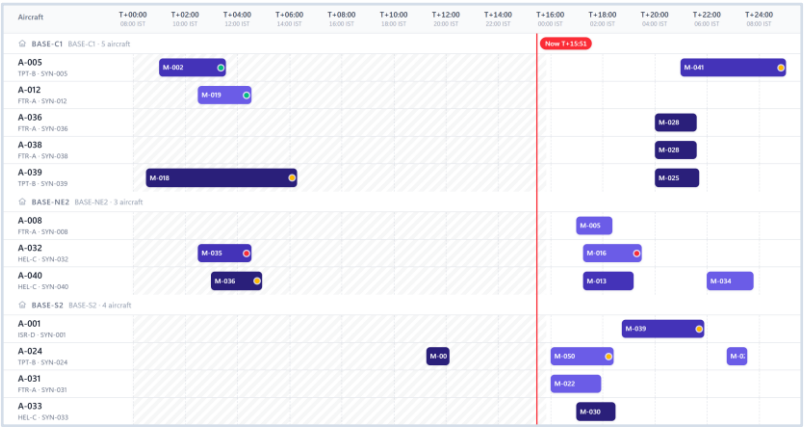
Google OR-Tools CP-SAT, with greedy and FIFO baselines; shapely + pyproj for routes and zones.

API and data

Python · FastAPI · Pydantic · SQLAlchemy. SQLite for development, PostgreSQL when deployed.

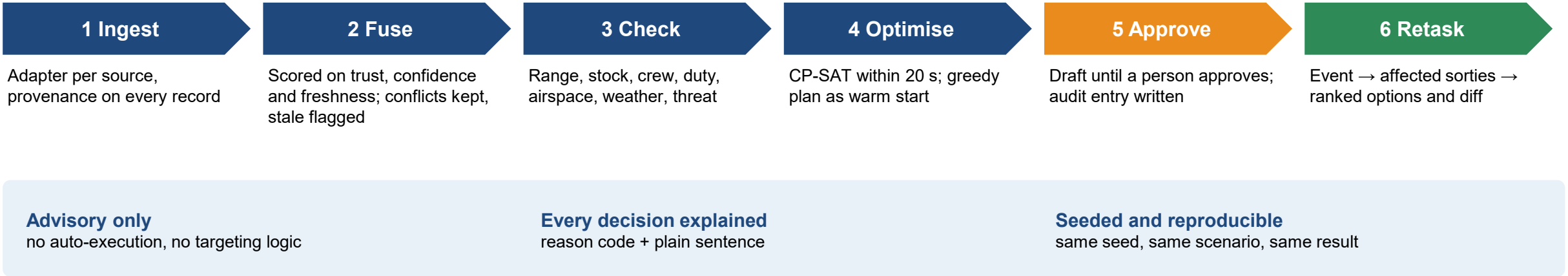
Assurance and hardware

pytest (385 tests) and an independent plan validator. No special hardware: a standard CPU.



Plan timeline: sorties by aircraft, locked = airborne

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)



Advisory only

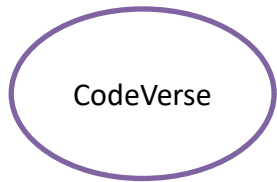
no auto-execution, no targeting logic

Every decision explained

reason code + plain sentence

Seeded and reproducible

same seed, same scenario, same result



FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea	Potential challenges and risks	Strategies for overcoming these challenges
• Technical: working end-to-end prototype and test suite • Compute: laptop CPU; plans bounded at 20 s; no GPU • Data: adapter layer; real feeds replace synthetic ones by configuration • Operational: advisory, so command authority is unchanged • Security: runs offline and air-gapped; open-source stack	• Data access: operational feeds are classified and fragmented • Data quality: late or conflicting reports can mislead a plan • Trust: planners will not act on a black box • Model fidelity: performance and risk figures are placeholders today • Adoption: doctrine, training, change management	• Accreditation first: on-premise, read-only adapters, role-based access • Freshness rules: stale and conflict flags; a human can pin a value • Explain and approve: reason codes, plan diff, audit trail • Real data: replace placeholders with service-approved figures • Shadow mode: run beside today's process before any reliance

Measured on the prototype (benchmark, not claimed)

20	17 of 20	11 of 20	0
seeded scenarios, 3 planners each	scenarios where CP-SAT beat greedy; worse in 0	plans proven optimal within the 20 s limit	violations in 60 validated plans

Where the optimiser times out it returns the best plan found; a fallback is always labelled as such. Source: benchmarks/RESULTS.md

Phased rollout and validation plan

Pilot · 0-3 mo	Trial · 4-9 mo	Expand · 10-18 mo	Scale · Yr 2+
1 planning cell, shadow mode, historical data	live read-only feeds, 2 cells, KPIs tracked	command-wide; ML models on real data	joint and HADR operations

Pilot KPIs: time to first approved plan, re-plan latency, priority-weighted coverage, share of proposals accepted

❖ Potential impact on the target audience

- P

Air-ops planners One fused picture and an explained draft plan instead of manual collation
- C

Duty controllers A disruption becomes affected sorties and ranked options, not a scramble
- K

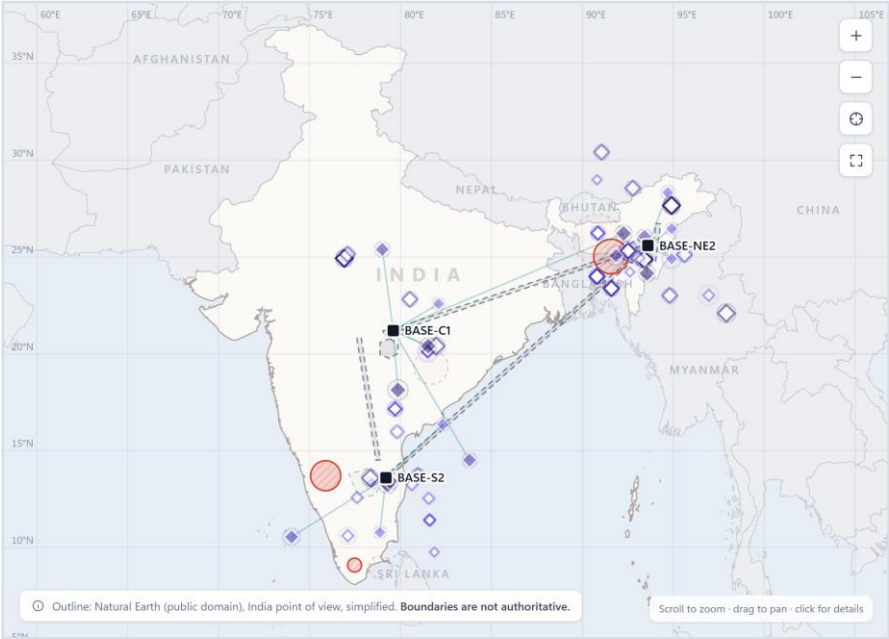
Commanders Coverage by priority, risk and a full audit trail at a glance
- H

HADR / NDMA relief Relief airlift allocated by priority and weather with the same engine
- M

Crew and maintenance Duty, rest and maintenance limits enforced at plan time, with the reason

❖ Benefits of the solution (social, economic, environmental, etc.)

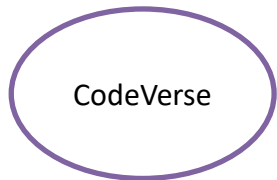
Social	Economic	Environmental	Governance
• Faster relief airlift in floods and cyclones • Less decision stress on planners	• Fewer wasted sorties and flying hours • Open-source stack, no licence cost	• Less unnecessary flying, so less fuel burned • Targeted re-plans, not full re-flights	• Human approval on every change • Append-only audit trail



Operating picture: bases, mission areas by priority, threat zones, airspace, sorties.
Outline: Natural Earth, not authoritative.

Pilot KPIs (baseline measured in shadow mode)

- Time to first approved plan vs the current process
- Re-plan latency after a disruption
- Priority-weighted coverage; sorties lost to late changes



RESEARCH AND REFERENCES



❖ Details / Links of the reference and research work

Problem and doctrine

- SIH 2026, Problem Statement SIH26250 (Ministry of Defence) · sih.gov.in
- US Joint Chiefs of Staff, JP 3-30 Joint Air Operations (joint air tasking cycle) · jcs.mil
- NDMA, National Disaster Management Plan: air support for HADR · ndma.gov.in

Human factors and automation

- Endsley (1995). Toward a theory of situation awareness in dynamic systems. Human Factors 37(1) · doi.org/10.1518/001872095779049543
- Parasuraman, Sheridan, Wickens (2000). A model for types and levels of human interaction with automation. IEEE Trans. SMC-A 30(3) · doi.org/10.1109/3468.844354

Optimisation and open data

- Google OR-Tools, CP-SAT solver (Apache-2.0) · developers.google.com/optimization
- Gopalan and Talluri (1998). Mathematical models in airline schedule planning: a survey. Annals of Operations Research 76
- Barnhart et al. (2003). Airline crew scheduling. Handbook of Transportation Science
- Natural Earth admin-0, India point of view · OurAirports · Open-Meteo elevation API (open data)

Data used in the prototype (transparency)

- **Real open data:** map outline (Natural Earth), alternate civil airfields (OurAirports), base elevations (Open-Meteo).
- **Synthetic data:** all bases, units, aircraft, crews, missions, threats, weather and events are generated from a seed and labelled SYNTHETIC in the app; performance figures are illustrative.
- **Measured results:** 20 seeded scenarios, 3 planners each; the script and CSV are in the repository, so no figure is typed in by hand.

SEE THE CODE · CHECK THE NUMBERS



Source code

github.com/Suyash2527/airpower-ops-optimiser